run_sjsdm_tuning_sequence.R

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`run_sjsdm_tuning_sequence`
Run the Shared sjSDM Tuning Sequence

Description

Orchestrates isolated unit and tier stores for exhaustive or staged tuning. Staged execution alternates one cumulative unit round with one tier-wide survivor target, then publishes the existing final tier artifact.

Usage

```
run_sjsdm_tuning_sequence(
  unit_pipeline = NULL,
  tuning_target_names = NULL,
  unit_store_suffixes = NULL,
  prebuild_interpolation = FALSE,
  fresh_run = FALSE,
  tuning_strategy = NULL,
  n_rounds = NULL,
  run_pipeline_function = run_pipeline,
  vec_allowed_profile_roles = base::c("main", "smoke"),
  vec_allowed_profile_statuses = "active",
  has_tuning_evidence_function = has_sjsdm_tuning_evidence,
  target_store = load_active_config_value("target_store"),
  load_metadata_function = load_targets_store_metadata,
  diagnose_endpoints_function = diagnose_sjsdm_cv_preparation_endpoints,
  read_target_function = targets::tar_read_raw,
  invalidate_target_function = targets::tar_invalidate,
 validate_tuning_targets = base::identical(run_pipeline_function, run_pipeline),
  cache_recovery_attempts = 8L,
  validate_calibration_function = validate_sjsdm_cv_production_calibration,
  unit_target_names = NULL
)
```

Arguments

`unit_pipeline`	Character scalar path to the unit pipeline script.
`tuning_target_names`	Non-empty character vector of public unit tuning-summary targets.
`unit_store_suffixes`	Optional character vector of isolated spatial-unit store suffixes. ‘NULL‘ runs one non-nested unit store, as used by temporal mapped pipelines.
`prebuild_interpolation`	Logical forwarded to [run_pipeline()] for unit executions.
`fresh_run`	Logical. When true, destroys unit stores on their first round and the tier store before its first aggregation. Later rounds always resume.
`tuning_strategy`	Character scalar, ‘"exhaustive"‘ or ‘"staged"‘.
`n_rounds`	Positive integer configured repeat/tuning-round count.
`run_pipeline_function`	Injectable pipeline runner. Defaults to [run_pipeline()].
`vec_allowed_profile_roles`	Profile roles forwarded to [run_pipeline()]. Production callers should retain ‘c("main", "smoke")‘; dedicated runners authorize other roles.
`vec_allowed_profile_statuses`	Profile statuses forwarded to [run_pipeline()]. Production callers should retain ‘"active"‘.
`has_tuning_evidence_function`	Injectable fail-closed evidence checker. Defaults to [has_sjsdm_tuning_evidence()].
`target_store`	Root targets-store path. Defaults to the active configuration value.
`load_metadata_function`	Injectable metadata loader used for legacy-cache recovery.
`diagnose_endpoints_function, read_target_function`	Injectable tuning-endpoint diagnostics and target reader.
`invalidate_target_function`	Injectable target invalidator used only for incompatible legacy caches.
`validate_tuning_targets`	Logical. Validate tuning artifacts after each production unit round.
`cache_recovery_attempts`	Positive integer maximum targeted legacy-cache repairs per unit and round.
`validate_calibration_function`	Injectable live calibration guard for main production profiles.
`unit_target_names`	Optional named per-unit tuning requests from verified preparation. The exact store-target pairs are persisted for resolution-aware tier pooling.

Value

Invisible ‘NULL‘. Pipeline stores contain the durable results. Unit-level failures are reported and skipped; successful units continue.

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